INCH-POUND
MIL-F-62422A(AT)
AMENDMENT 2
29 Jan 1993

MILITARY SPECIFICATION
FILTER, LASER HAZARD PROTECTION

This amendment forms a part of MIL-F-62422A(AT), dated 31 May 1991, and is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2 Add the following:

"Dwg. no. 12370300 Filter
Dwg. no. 12370324 Filter
Dwg. no. 7641866 Surface Quality Standards for Optical Elements"

PAGE 3

3.3 Delete and substitute:

"3.3 Construction. The filter shall be manufactured in accordance with drawing no. 12357797, 12357801, 12357857, 12370300, or 12370324."

PAGE 5

4.2 Add the following sentence: "The filter manufacturer shall be responsible for conducting certification tests."

4.2.1, lines 4 and 5: delete ", and witnessed by Government representatives"

PAGE 6

Add as new paragraph 4.3.2:

"4.3.2 First Article Samples. First article samples shall be three randomly selected filters of each type to be manufactured. The filters shall be selected from a full chamber run."

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Add as new paragraph 4.5.1.1:

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"4.5.1.1 Cosmetic - Filters shall show no characteristic(s) which are imperative or distractive to a clear, uniform field of view (FOV); i.e., blemishes or color variations which are discernible/distinguishable when viewed from the normal parameters of the FOV."

PAGE 11

4.6.2.1.2 Delete and substitute:

"4.6.2.1.2 Measurement - This test shall be conducted on the First Article samples. Detector responsivity calibrations are the same as those in Section 4.6.1.2. Each filter shall then be tested at a wavelength of 694 nm (see paragraph 3.5.1) using a linearly polarized laser with a collimated beam diameter of no less than 3 mm (FWHM). The energy distribution over the area of the laser pulse shall be a gaussian shape. Measurements of the noise level (flash tube pulse but no laser light to the detectors) without the filter in the optical train, shall be a reading recorded as an average of at least 100 pulses. The value of 100 pulses is used assuming that the minimum value achieved with the filter in the optical train is 1.5 orders of magnitude above the noise level. Similarly, readings 1 through 4 shall be values recorded as an average of at least 100 pulses (see 4.6.2.5 for exemption to 100 pulse requirement). All readings 4.6.2.1.2.2 through 4.6.2.1.2.4 shall be laser energy transmitted through different areas of the filter as shown in paragraph 4.6.6, Figure 2, except that only one center point shall be used. The laser energy incident on the filter shall be "P" polarized (minimum 100:1 ratio of "P" to "S"). The measurements shall be made such that the beam incident energy spreads in the fashion depicted by the rectangles in Figure 2."

4.6.2.1.2.1 Delete and substitute:

"4.6.2.1.2.1 Reading 1. The laser energy incident on the filter is measured. This can be performed directly or using a calibrated beamsplitter as in Figure 1 and then calculating the energy incident on the filter. If the reading is performed directly the laser energy incident on the filter must not drift by more than 5%. If a calibrated beamsplitter is used reading 1 shall be performed simultaneously with reading 2, reading 3, or reading 4 as required."

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4.6.2.2 Add the following sentence: "If a laser is used for optical density measurements see 4.6.2.5."

4.6.2.3 Delete and substitute:

"4.6.2.3. Optical density scan - Two filters per chamber load shall be tested for compliance with row A of Table I (see 3.5.1). One filter shall be chosen from the upper (or inner) area of the coating zone and one filter shall be chosen from the bottom (or outer) area of the coating zone. Optical density shall be measured in 0.5 degree increments (or less) and recorded with light incidence angles from 0 ± 1.5 degrees to 80 ± 1.5 degrees. The optical density scan shall be performed at the center of the filters. The energy incident on the filters

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shall be "P" polarized (minimum 100:1 of "P" to "S"). Filters shall be measured using a laser setup as in section 4.6.2.1.1 or a spectroradiometer setup as in section 4.6.4.1. Either system must be calibrated over the optical density range of 2 through 3.5 using NIST calibrated neutral density filters. In addition, either system shall be capable of measuring optical densities greater than 5.0 (noise limit > 5.0). Variations from either of the two test setups mentioned above must be submitted in a test plan and approved by the Contracting Officer. If a laser is used to perform the optical density scan, the optical density shall be measured for at least one pulse per angle."

* 4.6.2.3.1 Add as new paragraph:

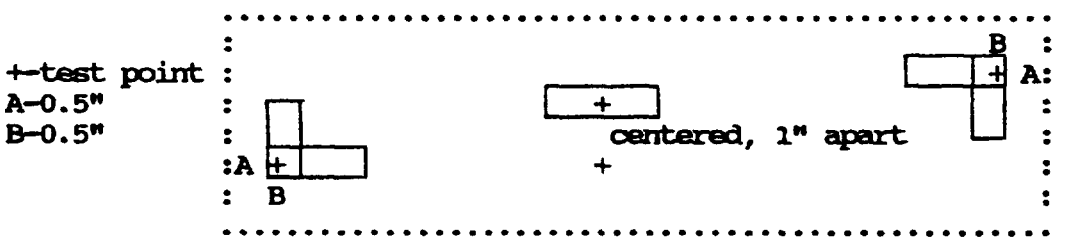
"4.6.2.3.1 Optical density for each angle shall be measured per 4.6.2.3 and recorded. The contractor shall plot optical density verses angle of incidence from 0 to 80 degrees. For any region of a valley where the optical density decreases by more than 0.7 and increases by more than 0.7 in 2 degrees or less and the minimum optical density is less than 4.4, the contractor shall locate and record the inflection point (angle) and the minimum optical density within that region by re-measuring the optical density within that region in 0.1 degree increments (or less)."

Add as new paragraph 4.6.2.5:

"4.6.2.5 Laser optical density measurements - Unless otherwise specified, all readings shall be recorded as an average of at least 100 pulses. The contractor may reduce the minimum number of pulses averaged from 100 to 10 if it is shown that all 10 single pulse optical density readings are within +/- 5% of the 10 pulse average optical density reading."

PAGE 14

Figure 2 Delete and substitute:



Note: A & B are distances from the edge

FIGURE 2

* 4.6.7, line 5: delete "Within one hour" and substitute "Within 15 minutes".

4.6.8 Add the following sentence: "Filters shall be evaluated using the surface quality standards for optical components drawing no. 7641866."

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4.6.9 Add the following sentences: "The energy density values specified in Table V are maximum energy density values. For a gaussian energy distribution the maximum energy density is twice the average energy density."

Add as new paragraphs 4.6.9.1, 4.6.9.1.1, 4.6.9.1.2:

"4.6.9.1 Calculations:

4.6.9.1.1 The average energy density is given by:

$$H_{avg} = \frac{Q}{A_2}$$

where Q is the total pulse energy and A_2 is the area bounded by the $1/e^2$ intensity points.

4.6.9.1.2 The maximum energy density is given by:

$$H_{max} = \frac{Q}{A_1} = 2\ H_{avg}$$

where Q is the total pulse energy and A_1 is the area bounded by the 1/e intensity points."

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